SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
    <input type="text" name="name">
    <input type="email" name="email">
    <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

Answers:
1.
The following semantic elements are missing:
<header> – for the symposium title.
<main> – for the main registration content.
<section> – for grouping registration details.
<footer> – for footer information.
<label> – to identify each form input clearly.
These elements make the webpage more structured and accessible.

2.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Symposium Registration</title>
</head>
<body>

<header>
   <h1>Online Symposium Registration</h1>
</header>

<main>
   <section>
      <h2>Student Registration</h2>

      <form action="/register" method="POST">

         <label for="name">Name:</label>
         <input type="text" id="name" name="name" required>

         <label for="email">Email:</label>
         <input type="email" id="email" name="email" required>

         <button type="submit">Submit</button>

      </form>
   </section>
</main>

<footer>
   <p>College Symposium 2026</p>
</footer>

</body>
</html>
```

3.

When the student clicks **Submit**, the browser sends a **POST request** to the /register URL.

Content-Type: application/x-www-form-urlencoded

The submitted data is sent in the request body, such as:

name=Ravi&email=ravi@example.com

Thus, the browser sends the student's registration details to the server using HTTP POST.

4.

The HTTP response cycle is:

1. Student fills in the form.
2. Student clicks **Submit**.
3. Browser sends a POST request to the server.
4. Server receives and processes the registration data.
5. Server sends an HTTP response.
6. Browser receives the response.
7. Browser displays the result to the student.

Example response:

HTTP/1.1 200 OK

The response may contain a registration-success message.

5.

The following improvements can be made:

- Add <label> for every input.
- Use required for necessary fields.
- Use appropriate input types such as email.
- Provide clear error messages.
- Use descriptive button text.
- Make the form keyboard accessible.
- Use semantic HTML elements.
- Make the webpage responsive for mobile devices.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

Answers:

1.

The given code is:

```
<div>
   <h1>Welcome
   <p>About us
</div>
```

The closing tags for <h1> and are missing.

Correct structure:

```
<div>
   <h1>Welcome</h1>
   <p>About us</p>
</div>
```

Thus, </h1> and must be added.

2.

Different browsers may use their HTML error-recovery mechanisms to correct the invalid structure.
This can cause:

- Different DOM structures.
- Different spacing or layout.
- Unexpected CSS behavior.
- Unexpected JavaScript behavior.

Therefore, properly written HTML should be used instead of depending on browser error correction.

3.

The webpage can use the following semantic elements:

- <header> – for the company heading.
- <main> – for the main webpage content.
- <section> – for the "About Us" content.
- <footer> – for footer information.

Example:

```
<header>
   <h1>Welcome</h1>
</header>

<main>
   <section>
      <h2>About Us</h2>
      <p>About our company.</p>
   </section>
</main>
```

4.

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Company</title>
</head>

<body>

<header>
    <h1>Welcome</h1>
</header>

<main>
    <section>
        <h2>About Us</h2>
        <p>About our company.</p>
    </section>
</main>

<footer>
    <p>© 2026 Company</p>
</footer>

</body>
</html>
```

This structure has proper nesting, closing tags, and semantic HTML5 elements.

5.

The following techniques can be used:

1. Use valid HTML5 syntax.
2. Properly close and nest elements.
3. Follow web standards.
4. Validate HTML and CSS.
5. Test the website in different browsers.
6. Use responsive design.
7. Avoid outdated or browser-specific features.
8. Test the webpage on different screen sizes.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
    <input type="text" name="username">
    <input type="password" name="password">
    <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Answers:

1.

The main issues are:

- The button uses type="button" instead of type="submit".
- GET is being used for login credentials.
- Labels are missing for the input fields.
- Input validation such as required is missing.
- HTTPS should be used for secure transmission.

2.

The HTTP request is not triggered because the button is:

```
<button type="button">Login</button>
```

type="button" does not submit the form.

It should be:

```
<button type="submit">Login</button>
```

The submit type tells the browser to submit the form to the URL specified in the action attribute.

3.

The GET method sends form data as part of the URL.

For example:

```
/submit?username=student&password=12345
```

This is not suitable for login because the password can appear in the URL and may be stored in browser history or logs.

Therefore, POST should be used for submitting login credentials.

4.

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Login</title>
</head>

<body>

<main>
    <section>
        <h1>Login</h1>

        <form action="/submit" method="POST">

            <label for="username">Username:</label>
            <input type="text" id="username" name="username" required>

            <label for="password">Password:</label>
            <input type="password" id="password" name="password" required>

            <button type="submit">Login</button>

        </form>
    </section>
</main>

</body>
</html>
```

The important corrections are:

- GET → POST

- Added required validation.

5.

The following improvements should be used:

1. Use **HTTPS** instead of HTTP.
2. Use **POST** for login credentials.
3. Validate input on the server.
4. Never store passwords as plain text.
5. Use secure password hashing.
6. Use secure session management.
7. Protect the application against common attacks such as SQL injection and XSS.
8. Do not expose passwords in URLs or error messages.

Code of Conduct:

*I **Kandath Haneesh*****(192324084)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kandath Haneesh

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand